PREDICTIVE ANALYSIS FOR CHURN MANAGEMENT AND CUSTOMER RETENTION

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Abstract-

Customer churn is a major issue in the Telecom industry. This paper uses five machine learning classifiers-Naive Bayes, Logistic Regression, KNN classifier, SVM classifier and Random forest on the IBM telco churn dataset to predict the customer churn.

The dataset was preprocessed by handling missing values, converting datatypes, removing duplicates, applying label encoding to binary valued columns and one-hot encoding/ dummy encoding to multi valued columns. To address class imbalance, synthetic minority oversampling technique (SMOTE) and ADAPTIVE Synthetic technique (ADASYN) were applied. Models were evaluated using accuracy, recall, precision, F1-score and AUC-ROC score.

Random Forest can be seen as the best-performing model for churn prediction after class balancing with SMOTE and ADASYN. It achieves the highest overall accuracy (77%), strong weighted F1-score (0.78), and a high AUC (0.82), indicating excellent distinguishability between churners and non-churners . It can also be seen that Logistic Regression leads on raw accuracy (82%) in an unbalanced setting. SVM achieves slightly higher recall for the minority class, Random Forest provides a more balanced and robust performance across all metrics, making it the practical choice.

This study shows that machine learning can effectively help companies to predict those customers who are at risk of churning and take necessary actions in time to prevent any loss of revenue.

I. INTRODUCTION

Customer churn is when a customer leaves a company's services. In the telecom industry, churn is a major problem because they lose money, and moreover, the cost of acquiring a new customer is more than retaining the existing one. Machine learning can help by analyzing customer data and predicting which customers are likely to churn before they actually leave, giving the company ample time to take action. This paper compares five machine learning classifiers- KNN, Naive Bayes, Random forest, SVM classifier and logistic regression on IBM telco customer churn dataset to identify the best performing model.

The complete implementation is available at:

https://github.com/Erikamediratta/Telecom-churn-prediction

II LITERATURE REVIEW

In the telecom sector, machine learning has been extensively studied to forecast client attrition. To determine the best method for churn prediction, numerous research have used and contrasted various classification algorithms on telecom datasets.

Wagh et al. (2024) [1]

Wagh et al. used machine learning classification algorithms to analyze customer attrition prediction in the telecom industry. Random Forest, KNN, and Decision Tree classifiers were applied to the IBM Telco Customer Churn dataset, which had 7043 records. Additionally, the authors used SMOTE and ENN approaches to alleviate class imbalance. After balancing, Random Forest outperformed other classifiers by achieving the maximum accuracy of 99%. The study found that managing class imbalance is essential for enhancing model performance and that Random Forest is the most dependable model for churn prediction. This outcome is in line with the current study's findings, which showed that Random Forest performed the best after using SMOTE and ADASYN.

Yuvalı et al. (2022) [2]

used two separate medical datasets to examine the classification performance of six machine learning algorithms, including KNN, SVM, Random Forest, Naive Bayes, Logistic Regression, and ANN. The concept of evaluating different classifiers using AUC and ROC curves is directly relevant to the current study, even though the study's focus was on coronary artery disease identification rather than telecom churn. The results demonstrated that KNN was the lowest classifier, with AUC scores varying from 0.47 to 0.52, while Random Forest consistently produced good performance throughout all experimental stages. In the current study, KNN only achieved an AUC of 0.50 on the high dimensional telecom dataset, demonstrating the trend of KNN underperforming due to high dimensionality.

IBM Telco Dataset [3] (2018)

The IBM Telco Customer Churn dataset included in this research is a benchmark dataset that is accessible to the general public on Kaggle. It has 7043 customer records with 21 attributes that include account details, subscribing services, and consumer demographics. The Churn column, which shows whether a customer has stopped using the service, is the goal variable. This dataset is appropriate for comparing findings from other studies because it has been extensively utilized in churn prediction research.

Pedregosa and associates (2011) [4]

An open-source Python machine learning framework called Scikit-learn offers effective implementations of a number of categorization techniques. Scikit-learn, a comprehensive Python machine learning package with standardized interfaces for model training, evaluation, and preprocessing, was first presented by Pedregosa et al. All five classifiers and assessment metrics, such as accuracy, precision, recall, F1-score, and AUC-ROC, were implemented in this work using Scikit-learn.

Few studies have thoroughly evaluated five classifiers with and without class balancing strategies like SMOTE and ADASYN while also adding explainability methods like SHAP and LIME, despite the fact that previous research has examined individual classifiers for churn prediction. This study fills this void.

III THEORETICAL BACKGROUND

We used five machine learning classifiers as follows:

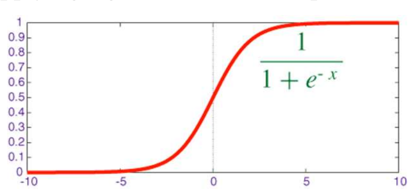
1. LOGISTIC REGRESSION-

It is a supervised machine learning algorithm that is used for classification problems, works better if there are just two classes(binary), for example- "Will a student pass or fail in the exam", email spam detection etc.

Works by finding the linear combination

Z=wx+b

Applying sigmoid function for probability,



This creates a decision boundary

P(y=1|x)>=0.5 class=1

P(y=1|x) <0.5 class=0

Sigmoid function allows use of log-loss(cross-entropy) instead of MSE (mean squared error)

2. SUPPORT VECTOR MACHINE(SVM) CLASSIFIER-

It is a supervised machine learning algorithm that is used for both classification and regression problems, It works well for high dimensional data, and its aim is to find the decision boundary that helps separate two classes, and maximize the margin between two classes (distance between the hyperplane and the closest points from both the classes(support vectors))

Hyperplanes- It is the decision boundary that separates different classes, SVM aims to find the "optimal" hyperplane , Interpretating the points on one side of the hyperplane belongs to class 1 and the other side belongs to class 0. Larger margin means better generalization.

Mathematically,

Wx+b=0 where:

W-weight vector (normal to hyperplane),

x-input vector,

b-bias/intercept

Margin=2/||w||

The C parameter controls the tradeoff:

Low C- allows for more misclassification, a more generalized boundary.

High C-models try to classify everything correctly, which increases the risk of overfitting.

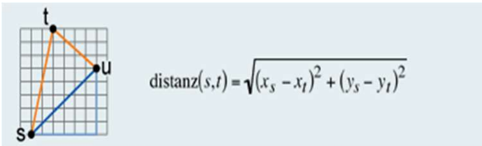
3. KNN CLASSIFIER-

It is an instance based, non-parametric machine learning algorithm that stores all training data and by finding the k closest training examples, classifies a new point.

It works as follows:

1. Selecting number of neighbors(k) and distance metric (mostly Euclidean)
2. Computing distance to all training points,

Euclidean distance=



3.Choosing the k training points with smallest distance and predicting by majority vote (classification) or mean (regression)

StandardScaler is applied before training, so as to prevent features with large ranges to dominate over the features with smaller ranges and making them irrelevant.

Curse of dimensionality-KNN algorithm relies on distance between points to find the nearest neighbors, as the features (dimensions) increases, all points become almost equidistant from each other which makes the concept of nearest neighbors irrelevant in high dimensional data.

Choice of k matters as small value of k (k=1), model is too sensitive to noise resulting in overfitting, larger k means too many nearest neighbors, resulting in underfitting, optimal k is found using cross-validation.

4.NAIVE BAYES CLASSIFIER-

It is a probabilistic machine learning algorithm used for classification based on bayes theorem. It is called naive because it assumes all features are independent.

Bayes theorem is as follows:

$$P(C|X)=P(X|C) \times P(C) / P(X)$$

Where:

P(C|X): Probability of class C given feature X

P(X|C): Probability of feature X given class C

P(X): Probability of feature X

P(C): probability of class C

Various types such as Gaussian Naive bayes which assumes features are continuous and follow a normal distribution.

5.RANDOM FOREST

It is an ensemble-based machine learning algorithm that makes multiple decision trees, instead of one single tree, for example if the dataset has 100 rows then we'll make 10 different datasets of 100 rows, then combining the predictions to get the best prediction by majority vote(classification) or averaging(regression). It uses bagging, reduces variance and is suitable for unstable models like decision trees. Random forests take random features at each split It might be slower than a single decision tree and needs more memory.

IV. METHODOLOGY

Dataset

The study uses the IBM telco customer churn dataset available publicly on Kaggle. The dataset has 7043 rows and 21 columns that include phone service, internet Service, gender, contract, total charges and more. The target variable is 'Churn' which indicates whether the customer has left the telecom services or stayed, represented by Yes or No making this a binary classification problem

Data preprocessing

The dataset required several preprocessing steps before model training, The TotalCharges column was found to be of object datatype despite containing numeric values due to the 11 blank spaces, so these were converted to NaN using pd.to_numeric() and were eventually dropped resulting in 7032 rows. The customerID column was dropped because it contained no predictive value. Duplicate records were found using df.duplicated().sum() ,(22 records), so they were dropped using df.drop_duplicates(), resulting in 7010 rows and 20 columns . Then encoding was applied, label encoding for two values columns such as gender, Churn and Phone Service . One hot-encoding was applied to multi-valued columns. drop_first=True was done because for example in column like Contract with 3 values(Month-to-Month, One year, Two year), one-hot-encoding will make 3 new columns, but we need only 2, the third one is automatically known to be 0 or 1 depending upon the other two columns ,this avoids multicollinearity, redundant columns that can confuse the model.

Train-test Split

The dataset was divided into features(X) and target(Y), X contains all columns except Churn and Y contains only the Churn column. The data was split into 80:20 ratio (80% training data and 20% testing data) using the train_test_split() from sklearn.model_selection. Random_state is taken as 42 to ensure reproducibility, means the same splitting ratio every time the code is run.

Evaluation metrics

Accuracy- It is the ratio of correct predictions to total predictions, however, accuracy alone as a metric can be misleading for imbalanced dataset like Churn dataset where most customers do not churn
Precision-It is a performance metric to identify how many predicted Yes were actually Yes
Recall-Performance metric used to identify how many of the actual positive cases were correctly predicted by the model
F1 Score- It is the harmonic mean of precision and recall
AUC-ROC-Performance metric used to evaluate how well a classification model distinguish between two classes, ROC is a curve plotting TPR(true positive rate) vs FPR(false positive rate) and AUC is the area under the curve . Higher AUC (close to 1) means model operates the classes in a better way.

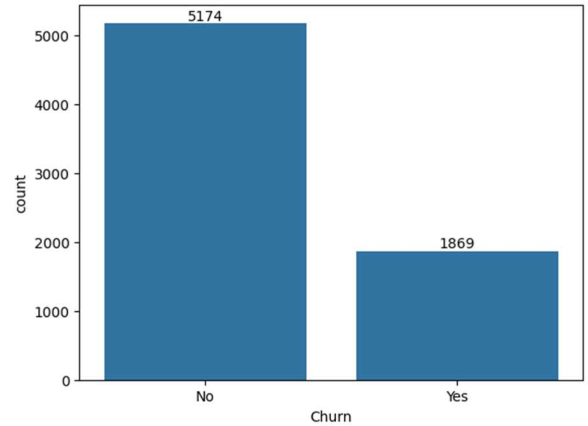
Issues in Classification

While solving classification problems, we might encounter problems such as,
Underfitting- The model is too simple to learn patterns, it has High training error as well as high testing error, along with high bias and low variance.
Overfitting- The model is too complex and too sensitive to training data; it learns patterns along with noise and outliers, it has high variance and low bias (Works phenomenal for training data but fails for testing data).
For this, we need a sweet spot that balances out (Bias-variance tradeoff)

Another problem that we can potentially face is class Imbalance problem, the IBM telco churn dataset suffers from class imbalance with 5174 non-churners (74%) and 1869 churners (26%). This imbalance can cause models to be biased towards the majority class, which can therefore result in misleading accuracy scores. To address this, smote was applied to training data to balance class distribution before model training.

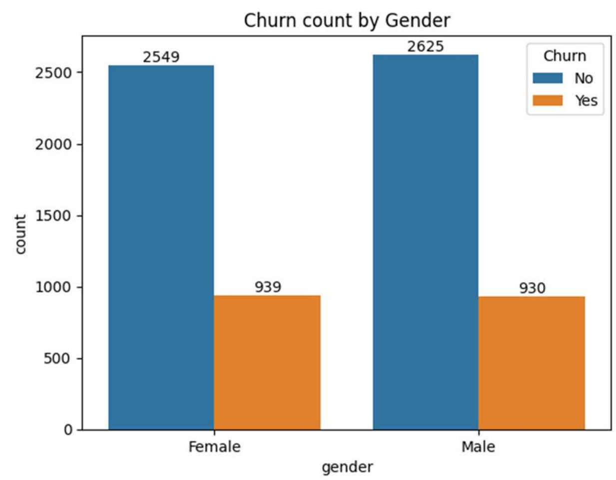
Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to understand the distribution and pattern in the dataset



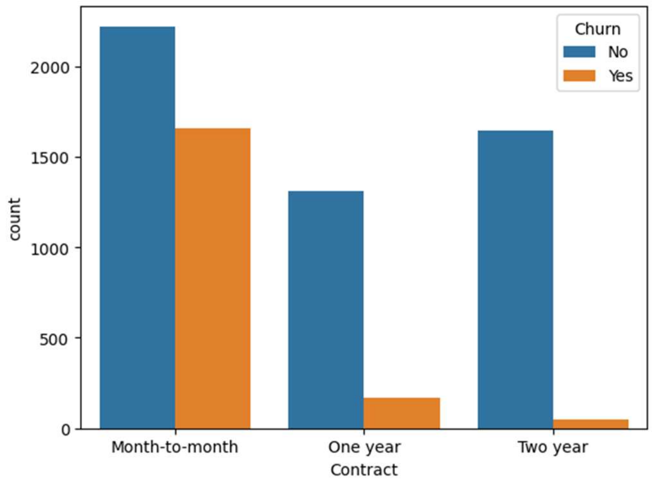
CHURN COUNTPLOT

It can be seen from the plot that the dataset is imbalanced with 5174 customers who have not churned out (active) and 1869 have churned.



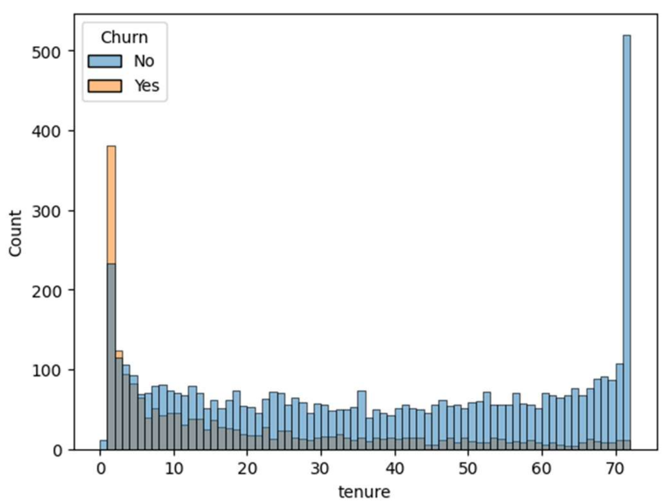
GENDER VS CHURN COUNTPLOT

It is visible that the churn is not influenced by the gender, indicating identical behavior across both males and females.



CONTRACT VS CHURN COUNTPLOT

It can be seen that the customers with monthly basis contract have churned out more then one year contract and two year contract.



TENURE VS CHURN

This histogram reveals that new customers (low tenure) are more likely to churn than long term customers, and the customer retention efforts must focus on newly acquired customers.

Model Explainability

Models like Random Forest and other ensemble algorithms are considered as "Black Box". Explainable AI techniques like SHAP (Shapley additive explanations) and LIME (Local interpretable model- agnostic explanations) helps understand which features are most influential which helps build trust and increases usability in real-world applications.

LIME- It explains individual predictions and explains why a specific case is predicted as positive or negative.

SHAP- It is based on game theory and explains global feature importance and how does a specific feature contributes to a prediction made.

V. RESULTS AND DISCUSSION

The Performance of the five machine learning classifiers- Logistic Regression, Random Forest, Naive Bayes, K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) is evaluated looking at Accuracy, precision, recall, F1-score and Area under the ROC curve (AUC) . The models were tested before and after applying synthetic minority oversampling technique (SMOTE) and Adaptive Synthetic Technique (ADASYN) to analyze the effect of class imbalance correction.

A. BEFORE SMOTE AND ADASYN :

The evaluation results before applying SMOTE are summarized as follows in table 1.

Model	Accuracy	Precision	F1	Recall	AUC
Logistic Regression	82%	0.82	0.82	0.82	0.85
Random Forest	81%	0.81	0.81	0.81	0.83
Naive Bayes	65%	0.82	0.65	0.67	0.82
KNN	77%	0.77	0.77	0.77	0.74
SVM	82%	0.81	0.82	0.81	0.80

Table 1

A. Individual model Analysis:

Logistic Regression -
Confusion Matrix: [[972, 109], [138, 183]]
Out of 321 churned customers, the model identified 183 correctly and 138 were missed, out of 1081 non-churned customers, 972 were correctly predicted and 109 were wrongly flagged. Logistic Regression has achieved the best results across all models with the highest accuracy (82%) with a good recall of 0.82, and good AUC (0.85) which distinguishes the churners well.

Naive Bayes-
Confusion Matrix: [[618, 463], [34, 287]]
Out of 321 churned customers, the model identified 287 correctly and 34 were missed, Out of 1081 non-churned customers, 618 were correctly predicted and 463 were wrongly flagged. Since it assumes all features are independent, this assumption is violated as MonthlyCharges and TotalCharges are highly dependent, so accuracy is poor (65%).

KNN-
Confusion Matrix: [[902, 179], [149, 172]]
Out of 321 churned customers, the model identified 172 correctly and 149 were missed, Out of 1081 non-churned customers, 902 were correctly predicted and 179 were wrongly flagged. KNN had a poor performance because there were too many columns after one-hot encoding and in KNN, we calculate distances, so they become meaningless, this is called curse of dimensionality

SVM-
Confusion Matrix: [[986, 95], [156, 165]]
Out of 321 churned customers, the model identified 165 correctly and 156 were missed, Out of 1081 non-churned customers, 986 were correctly predicted and 95 were wrongly flagged.
In comparison to Random Forest and Logistic Regression, SVM's AUC (0.80) indicates a moderate capacity to discriminate, despite its high accuracy of 82%. This could be because high-dimensional categorical encoding adds complexity and is sensitive to hyperparameter selection.

Random Forest-
Confusion Matrix: [[971, 110], [151, 170]
Out of 321 churned customers, the model identified 170 correctly and 151 were missed, out of 1081 non-churned customers, 971 were correctly predicted, and 110 were wrongly flagged. Although slightly lower in accuracy than logistic Regression, Random Forest is more robust to noise and outliers due to ts ensemble nature.

However, Random Forest may not perform well for small datasets because it randomly selects subset of features for each tree. In small datasets, this may lead to exclusion of the important features. Additionally, random forest builds deep decision trees, which may create splits for very few or rare observations. This will lead to exceptionally good performance on training data but will fail for testing data- classic overfitting problem.

This suggests that distance and margin-based algorithms are not very well suited for such high dimensional data without proper hyperparameter tuning.

B. AFTER SMOTE:

SMOTE (Synthetic minority oversampling technique) treats all minority points equally, it
does not care if some points are harder to classify for example boys are 90 and girls are 10, so SMOTE would make fake minority points (girls).

Before smote Churn
0 4072
1 1536

After smote Churn
0 4072
1 4072

The evaluation results after applying SMOTE to treat class imbalance are summarized in table 2.

Model	Accuracy	Precision	F1	Recall	AUC
Logistic Regression	76%	0.79	0.78	0.77	0.81
Random Forest	77%	0.78	0.78	0.77	0.82
Naive Bayes	61%	0.81	0.64	0.61	0.81
KNN	70%	0.77	0.72	0.70	0.76
SVM	69%	0.75	0.71	0.70	0.82

Table 2
The recall shown in the table 1, table 2 , table 3 are weighted recalls.

After applying SMOTE, It can also be seen that there is a slight reduction in accuracy across all models. This is because balancing the dataset can make classification a bit challenging compared to learning from an imbalanced dataset where the majority class dominates prediction.

it can be observed that recall for the churn class increased for the majority of models after SMOTE was applied. While Naive Bayes maintained a high recall of 0.89, Logistic Regression increased from 0.57 to 0.64, Random Forest from 0.53 to 0.58, KNN from 0.54 to 0.65, and SVM from 0.51 to 0.57. This indicates that even while overall accuracy marginally declined, SMOTE improved the models' ability to identify churn consumers.

This indicates that the models became better at identifying the churned customers which is more important since missing a churn customer is more costly than incorrectly flagging a non-churn customer.

Even though accuracy decreases, the improved minority class recall demonstrates that SMOTE helps models learn churn behavior better.

C. AFTER ADASYN

ADASYN (Adaptive synthetic technique) is another technique to handle class imbalance, and it focuses on the minority points that are harder to classify, for example, if girl A (surrounded by many boys) , girl B (surrounded by many girls), so, would give priority to girl A(harder to classify as surrounded by the majority group), creates more points for such harder ones, and fewer points for the easy ones.

Before ADASYN Churn
0 4072
1 1536

After ADASYN Churn
0 4072
1 4008

The evaluation results after applying ADASYN to treat class imbalance are summarized in table 3.

Model	Accuracy	Precision	F1	Recall	AUC
Logistic Regression	75%	0.79	0.77	0.76	0.80
Random Forest	77%	0.79	0.78	0.77	0.82
Naive Bayes	61%	0.81	0.64	0.61	0.80
KNN	69%	0.77	0.72	0.70	0.73
SVM	71%	0.81	0.73	0.71	0.82

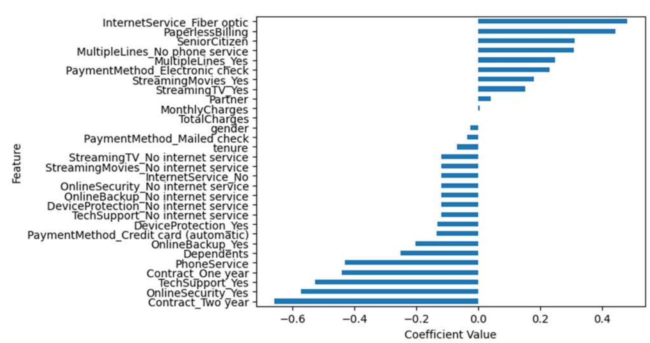
Table 3

After applying Adasyn, the best model is Random forest, Naive Bayes has high precision but very low recall which is not ideal for churn detection, SVM has good AUC and decent recall, but slightly lower F1 than random forest and so on. For churn prediction, prioritize recall+F1 over just accuracy. KNN's AUC slighlty declined after ADASYN(0.73) compared to SMOTE (0.76) . It is because ADASYN generates synthetic points in harder boundary regions and KNN's distance based nature makes it more sensitive to boundary synthetic samples.

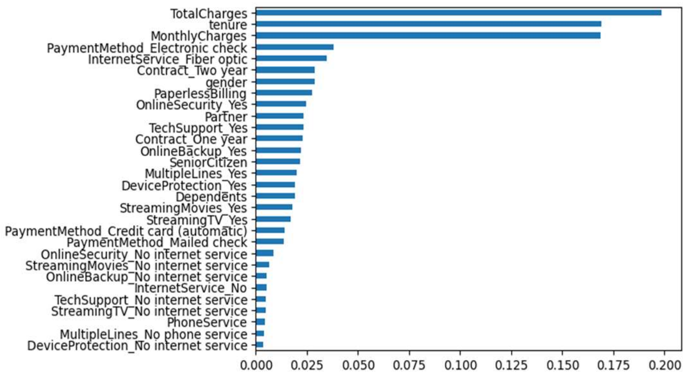
D. Overall Findings

Without Balancing: Logistic regression is slightly better- high accuracy, recall, F1 and AUC.
With Balancing (SMOTE or ADASYN) : Random Forest becomes more robust, means it can handle minority points better. Slightly higher AUC and F1.
For imbalanced datasets. Random forest + balancing (SMOTE/ ADASYN) is usually safer and more reliable, even if Logistic regression looked a little better unbalanced. This is because logistic Regression can overestimate performance when the dataset is imbalanced because it might just predict the majority class well.

E. Feature Analysis



From this plot, its visible that Internet service Fiber optic has the highest positive coefficient in logistic regression which makes sense as they pay more monthly charges and are more likely to churn if not satisfied.

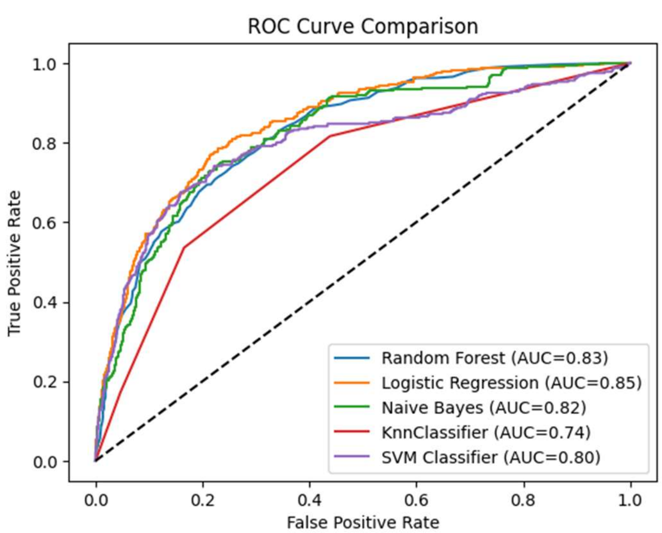


Feature importance- how much does a specific feature help in predicting whether the customer will churn or not.
From this graph, it can be seen that the most important features are in the order-TotalCharges, tenure, MonthlyCharges and so on.

F. ROC CURVE COMPARISON

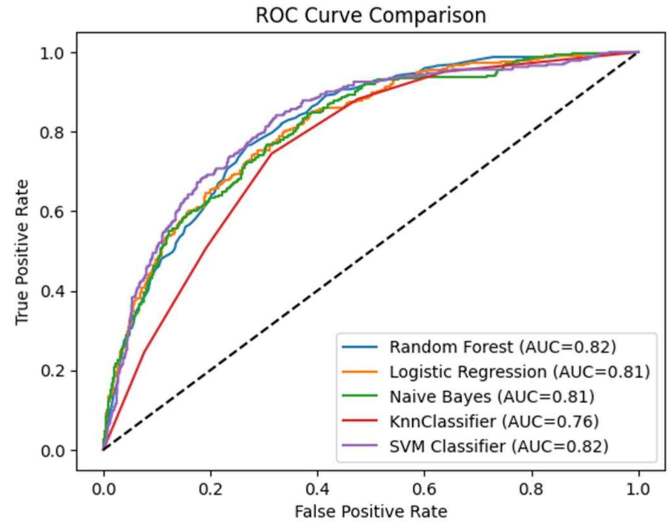
BEFORE SMOTE AND ADASYN

ROC curve comparison shows that Logistic regression achieved the best AUC (0.85) , followed by Random Forest (0.83). SVM and Naive Bayes showed moderate distinguishability, and KNN showed comparatively lower AUC.

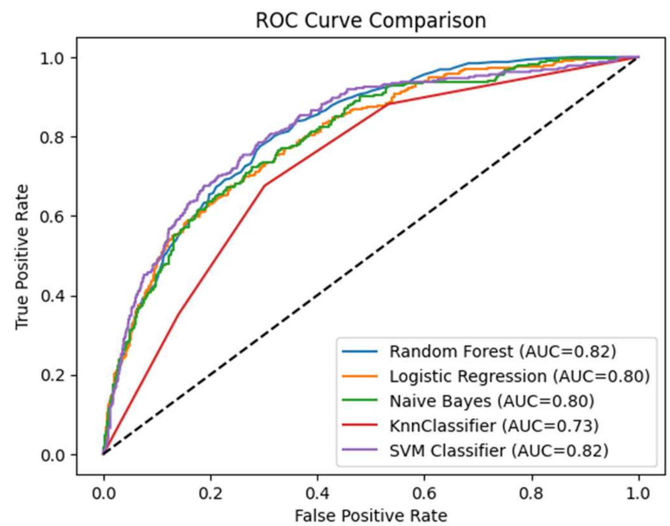


AFTER SMOTE

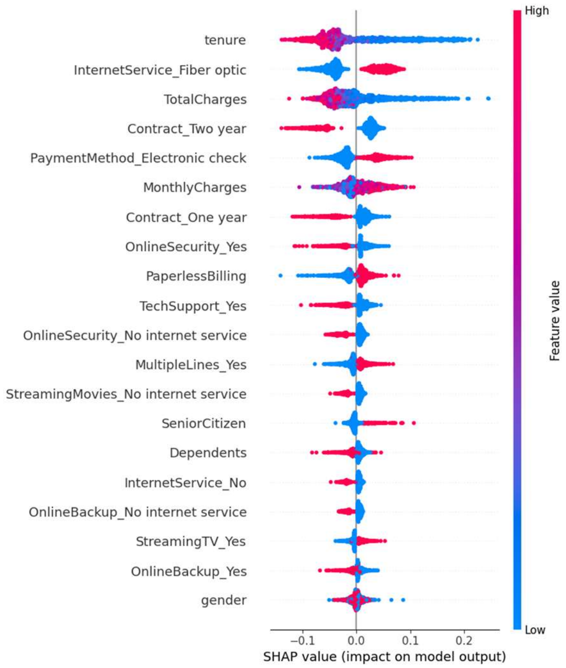
After applying SMOTE, Roc curves show that Random Forest and SVM achieved highest AUC (0.82) followed closely by Logistic Regression and Naive Bayes (0.81). KNN showed slight improvement (0.76)



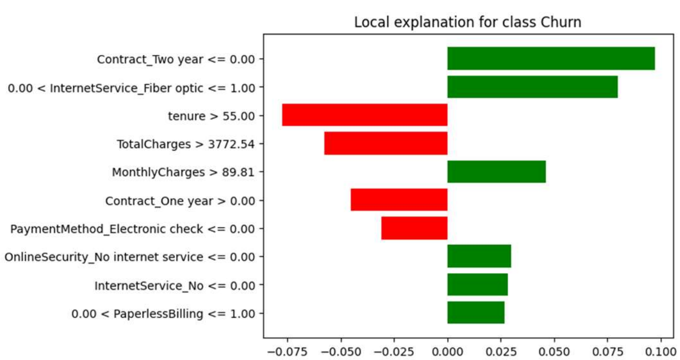
AFTER ADASYN



G. EXPLANAIBILITY ANALYSIS



The SHAP summary plot shows that tenure, internetService_fibre optic and TotalCharges are the most influential features. Blue dots on right explain that low feature value increases prediction whereas red dots on right explain high feature value increases prediction.



This LIME plot shows that high tenure and high total charges reduce churn probability, and absence of long-term contract slightly increases churn risk.

VI. CONCLUSION

Logistic Regression, Random Forest, Naive Bayes, K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) classifiers were implemented on the IBM Telco Customer Churn dataset to predict customer churn behavior.

Random forest after balancing (SMOTE/ADASYN) showed the best overall performance across all five models , indicating strong discrimination between churn and non-churned customers.

After applying SMOTE/ADASYN , although a slight reduction in overall accuracy was observed but the recall of the minority churn class improved across most models, indicating better identification of churn customers . This demonstrates the trade-off between overall accuracy and minority class detection when handling imbalanced datasets.

In future work, more machine learning models such as Decision Trees and Gradient Boosting methods can be added. Further improvements may also be achieved through hyperparameter tuning, feature selection, and advanced imbalance handling techniques to improve prediction robustness and model generalization.

VII. REFERENCES

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